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Iron-Catalyzed Three-Component Cyanoalkylsulfonylation of 2,3-Allenoic Acids, Sulfur Dioxide, and Cycloketone Oxime Esters: Access to Cyanoalkylsulfonylated Butenolides

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Abstract: An iron-catalyzed three-component cyanoalkylsulfonylation of 2,3-allenoic acids, $K_2S_2O_5$, and the ring-opening of cyclobutanone oxime esters is described. The radical tandem cyclization route allows access to diverse cyanoalkylsulfonylated butenolides in moderate to good yields under mild conditions. Moreover, the products are further converted, offering the corresponding derivatives.

Keywords: Butenolides; Three-component; Cyanoalkylsulfonylation; Iron-catalyzed; Derivatization

Introduction

Furan-2(5*H*)-ones represent a prominent class of structural scaffolds widely found in natural products and biologically active compounds (Figure 1).[1] Allenoic acids have been intensively employed as synthons to obtain furan-2(5*H*)-one derivatives, owing to their high reactivity and easy preparation.[2] For example, Ma and co-workers initially developed palladium-catalyzed coupling cyclization of 2,3-allenoic acids with allenes,[3] alkynes,[4] alkenes,[5] or aryl iodides[6] to afford polysubstituted butenolides (Scheme 1a). Afterward, Ma[7] and Qing[8] reported the copper-catalyzed free radical cyclization of 2,3-allenoic acids with various free radical donors to form various furan-2(5*H*)-one compounds. In addition, Wu[9] and co-workers reported the synthesis of arylsulfonylated butenolides *via* a copper(II)-catalyzed three-component reaction (Scheme 1b). Despite the successful implementation of these synthetic methods, the development of novel types of catalysts and multicomponent reactions to obtain complex furan-2(5*H*)-ones remains highly desirable.

Cyanoalkylsulfonyl groups are valuable organic building blocks for the synthesis of bioactive molecules.[10] In addition, they can also be easily transformed into other useful functional groups.[11] In recent years, various synthetic approaches have been established for the introduction of cyanoalkylsulfonyl groups. In 2008, Felton[12] reported the electrochemical reduction of alkyl vinyl sulfones with acetonitrile to obtain cyanoalkylsulfonyl groups. Later, Landais and coworkers[13] demonstrated the three-component cyanoalkylsulfonylation of α-bromosulfone, olefin, and *p*-tosyl cyanide. Moreover, Chen[14] reported the copper-catalyzed cross-coupling of oxime esters and sulfinates for the synthesis of cyanoalkylsulfonyl groups. Although some examples[15] of the synthesis of

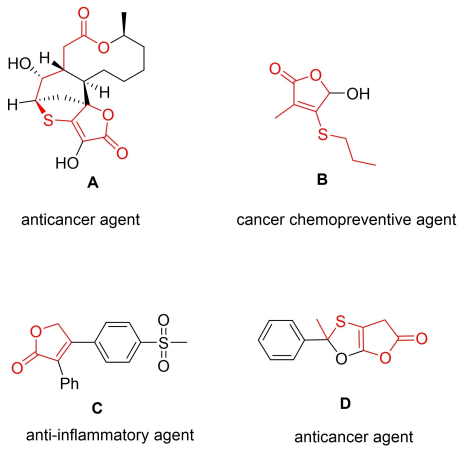
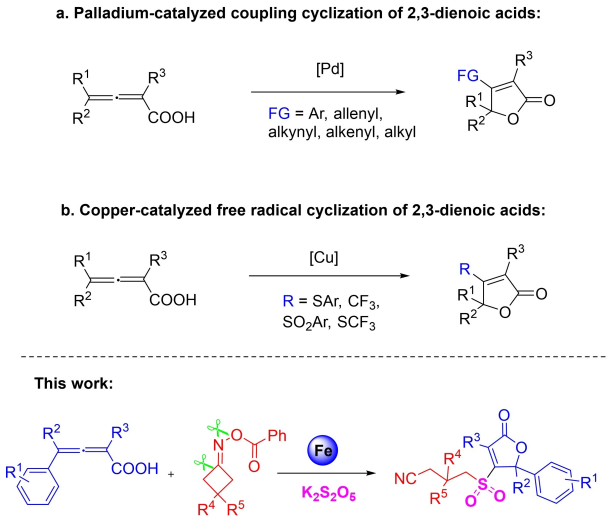


Figure 1. Biologically active compounds containing furan-2(5*H*)-ones.

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Scheme 1. Synthesis of various butenolides.

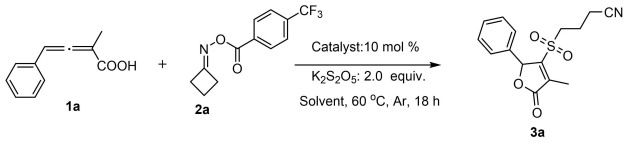
cyanoalkylsulfonyl groups using cyclobutanone oxime esters[16] and $K_2S_2O_5$[17] as cyanoalkyl and sulfonyl group sources,[18] respectively, have been reported, synthesis methods of cyanoalkylsulfonylated butenolides are not very common. Based on our previous work on radical synthesis,[19] herein we report the iron-catalyzed three-component cyanoalkylsulfonylation of 2,3-allenoic acids and the ring opening of *O*-acyl oximes with the insertion of sulfur dioxide.

Results and Discussion

First, a model reaction between 2-methyl-4-phenyl-2,3-allenoic acid (**1a**), $K_2S_2O_5$, and cyclobutanone *O*-(4-(trifluoromethyl)benzoyl) oxime (**2a**) was carried out in the presence of catalytic $Cu(OAc)_2$ under Ar atmosphere at 60 °C for 18 h, and the cyanoalkylsulfonylated product 3a was isolated in 18% yield (Table 1, entry 1). Among the various screened catalysts (entries 1–9), $FeCl_2$ proved to be most suitable one, providing **3a** in 65% yield (entry 3). However, lower yields (0–25%) were obtained when other solvents, such as MeCN, tetrahydrofuran (THF), and 1,4-dioxane were employed (entries 10–12). Next, screening of sulfur dioxide surrogates revealed that $K_2S_2O_5$ was still the most efficient reagent (entries 13 and 14). In addition, changing the reaction temperature and reducing the reaction time resulted in lower yields (entries 15–17). Interestingly, a 75% yield of **3a** was achieved upon extending the reaction time to 24 h (entry 18). Therefore, the optimal conditions were determined as follows: **1a** (0.2 mmol), **2a** (0.3 mmol), $K_2S_2O_5$ (0.4 mmol), $FeCl_2$ (10 mol.%), and DCE (2 mL), under Ar atmosphere at 60 °C for 24 h.

Based on the above optimized conditions, we next explored the substrate cyanoalkylsulfonylation reaction

Table 1. Optimization of reaction conditions.[a]

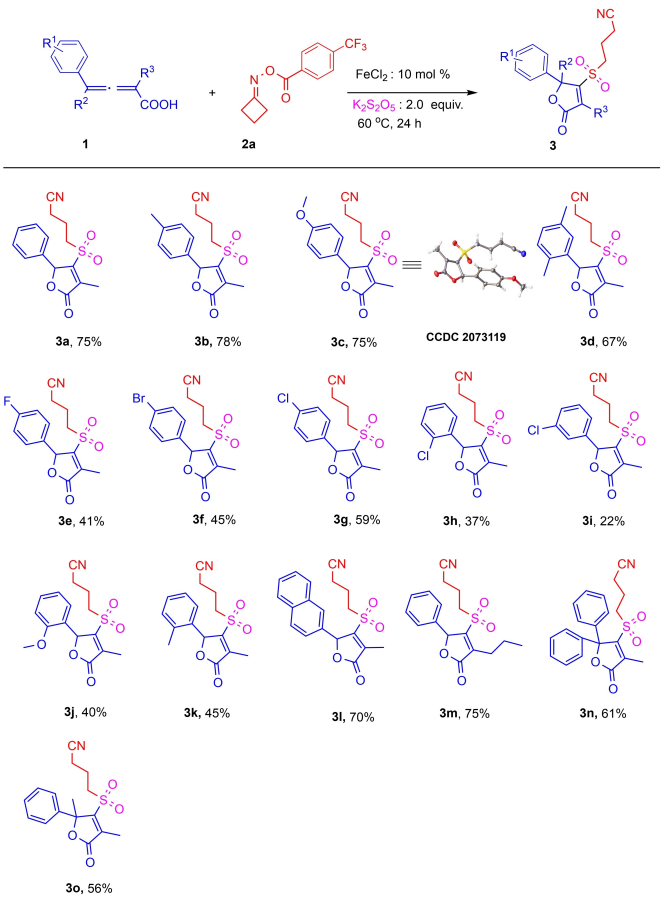


Entry	Catalyst	Solvent	Yield (%)[b]
1	$Cu(OAc)_2$	DCE	18
2	CuOAc	DCE	trace
3	$FeCl_2$	DCE	65
4	$Ni(acac)_2$	DCE	0
5	$Cu(CH_3CN)_4BF_4$	DCE	20
6	$Fe(OTf)_2$	DCE	35
7	$Fe(OTf)_3$	DCE	50
8	$FeCl_3$	DCE	54
9	$Fe(OAc)_2$	DCE	45
10	$FeCl_2$	CH_3CN	25
11	$FeCl_2$	THF	trace
12	$FeCl_2$	1,4-dioxane	8
13[c]	$FeCl_2$	DCE	42
14[d]	$FeCl_2$	DCE	10
15[e]	$FeCl_2$	DCE	42
16[f]	$FeCl_2$	DCE	60
17[g]	$FeCl_2$	DCE	55
18[h]	**$FeCl_2$**	**DCE**	**75**

[a] Reaction conditions: **1a** (0.2 mmol), **2a** (0.3 mmol), catalyst (10 mol%), $K_2S_2O_5$ (2 equiv.), and DCE (2 mL) at 60 °C for 18 h under Ar atmosphere.
[b] Isolated yield.
[c] $Na_2S_2O_5$.
[d] DABSO.
[e] 50 °C.
[f] 70 °C.
[g] 12 h.
[h] 24 h.

with different allenes (Scheme 2). We mainly considered 2,3-allenoic acids bearing the R^1 substituent group on the aromatic ring. Different substituents containing electron-donating groups such as methyl and methoxy groups in *para* position or bis-methyl substituents of the aromatic ring reacted smoothly and with high efficiency, delivering the cyclization products **3a–3d** in 67–78% yields. Halogen substituents at the *para*, *ortho*, or *meta* positions of the aromatic ring were also well tolerated, affording the corresponding compounds **3e–3i** in 22–59% yields. However, substrate **1**, bearing methyl and methoxy groups at the *ortho* position, delivered the desired products **3j** and **3k** in relatively low yields compared to electron-donating groups at the *para* position, presumably due to steric hindrance effects. In addition, when the aromatic ring was replaced by a naphthalene unit, an unexpected cyclization product (**3l**) was formed in 70% yield. Interestingly, this reaction type could also be utilized for the substrate with *n*-propyl as R^3 substituent (**3m**).

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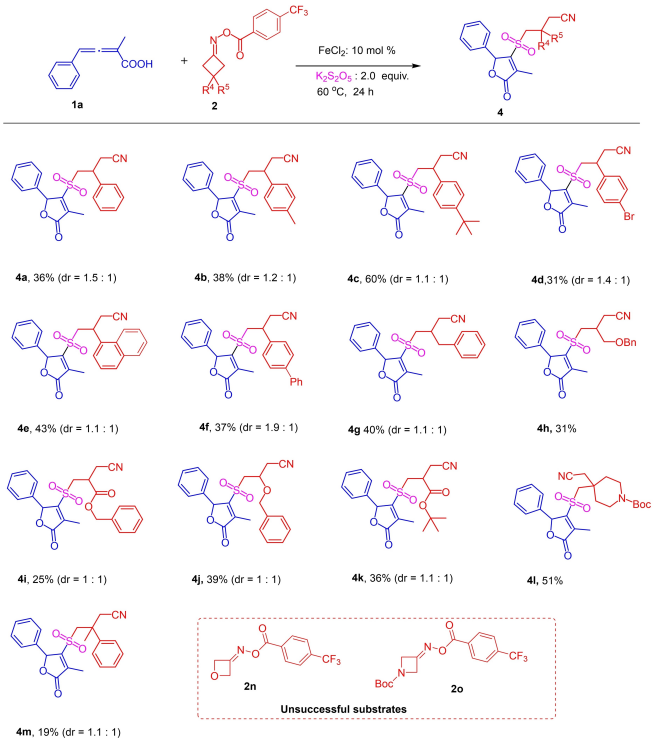


Scheme 2. Reaction of 2,3-Allenoic acids **1** with cyclobutanone *O*-(4-(trifluoromethyl)benzoyl) oxime **2a**. [a]Reaction conditions A: **1** (0.2 mmol), **2a** (0.3 mmol), $FeCl_2$ (10 mol%), $K_2S_2O_5$ (2 equiv.), and DCE (2 mL) at 60 °C for 24 h under Ar atmosphere. [b] Isolated yields.

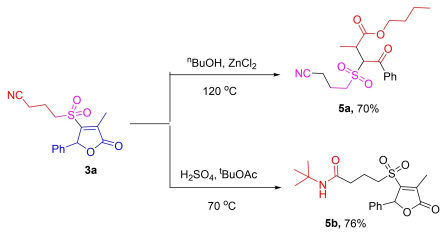
Similarly, both Ph and Me groups were effective substituents at the R^2 position (**3n** and **3o**). The crystal structure of compound **3c** was confirmed by X-ray crystallographic analysis (see Supporting Information for details).[20]

To further extend the versatility of this radical cascade cyclization reaction, we evaluated a series of 3-substituted cyclobutanone oxime esters **2** (Scheme 3). As expected, various functional groups including aryl, benzyl, ester, and benzyloxy moieties were compatible, producing the desired products **4a**–**4k** (25–60%). Importantly, 3-disubstituted compounds also reacted smoothly to afford the **4l** and **4m** compounds in 51% and 19% yields. However, other cycloketone oximes **2n** and **2o** failed to form the desired product.

We synthesized derivatives of sulfonylated product **3a**, to demonstrate its synthetic value (Scheme 4). For instance, **3a** could undergo lactone ring opening and butanolysis to yield product **5a** in the presence of *n*-butanol and $ZnCl_2$. Furthermore, the cyano group of



Scheme 3. Reaction of 2-methyl-4-phenyl-2,3-allenoic acid **1a** with cyclobutanone oxime esters derivatives **2**. [a] Reaction conditions A: **1a** (0.2 mmol), **2** (0.3 mmol), $FeCl_2$ (10 mol%), $K_2S_2O_5$ (2 equiv.), and DCE (2 mL) at 60 °C for 24 h under Ar atmosphere. [b] Isolated yields.

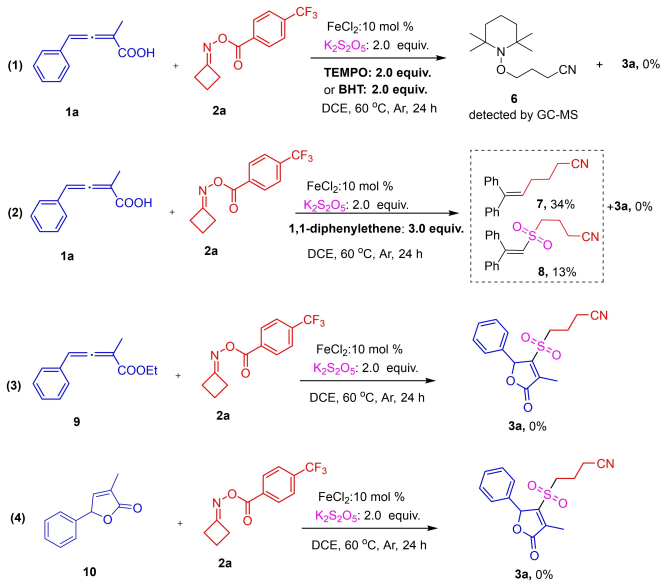


Scheme 4. Transformation of butenolides.

3a can easily be transformed to provide the *N*-*tert*-butyl-substituted amide **5b** through a Ritter reaction.

A series of control experiments were conducted to determine the detailed reaction mechanism (Scheme 5). Although product **3a** was not obtained in the presence of the 1,1-diphenylethene (DPE), TEMPO, and BHT radical scavengers, compound **6** was detected by GC-MS. Furthermore, compounds **7** and **8** could be isolated in 34% and 13% yield respectively when DPE was added to a sulfonylation reaction mixture. These experimental results indicate the occurrence of a radical process in this reaction. On the other hand, no reaction occurred when the 2,3-allenoate **9** replaced allenoic acid, proving the crucial role of allenoic acid as a directing group for the cyanoalkylsulfonylation

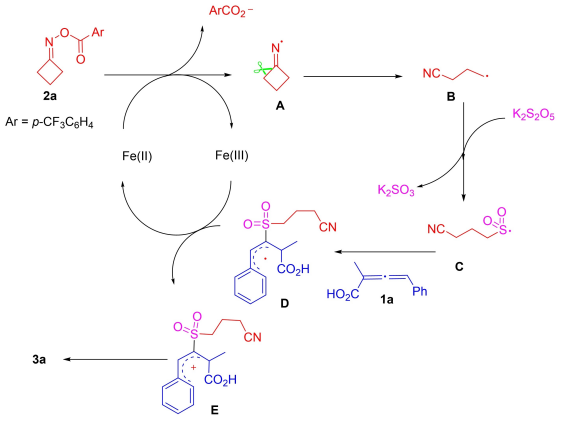
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Scheme 5. Control experiments.

reaction. In addition, the reaction of **10** and **2 a** did not yield the desired product **3 a** under standard conditions, suggesting that 2,3-allenoic acids may not first undergo intramolecular cyclization to form **10**, and then reacted with sulfonyl radical.

Based on the control experiments discussed above as well as previous studies,[7,8,15a–c] we propose a possible free radical pathway for the present cascade cyclization. (Scheme 6). Initially, SET reduction of the cycloketone oxime ester **2 a** catalyzed by Fe(II) leads to the cleavage of N–O bonds, forming the iminyl radical intermediate **A**. Subsequently, the C–C bond cleavage of intermediate **A** yields the alkyl radical species **B**. Next, sulfur dioxide, originated from $K_2S_2O_5$, attacks the intermediate **B** to give sulfonyl radical **C**, followed by addition to 2,3-allenoic acid **1 a** to form the intermediate **D**. Under Fe(III) catalysis, the allylic cation **E** is generated from the oxidation of radical species **D**, while Fe(II) is regenerated. Finally, the intramolecular nucleophilic attack of intermediate **D** yields the cyclization product **3 a**.



Scheme 6. Proposed reaction mechanism.

Conclusion

In conclusion, we have successfully developed an iron-catalyzed cyanoalkylsulfonylation of 2,3-allenoic acids, leading to a variety of cyanoalkylsulfonylated butenolides under mild conditions. The use of $K_2S_2O_5$ as sulfur dioxide surrogate renders this approach both safer and more environmentally friendly than other synthetic methods. Moreover, further transformations of the cyanoalkylsulfonylated butenolides provided the corresponding derivatives. Most of these compounds were previously unknown; additional experiments evaluating the further applicability of this protocol are currently being performed in our laboratory.

Experimental Section

General Procedure for the Synthesis of Cyanoalkylsulfonylated Butenolides

A mixture of 2,3-allenoic acids 1 (1 equiv., 0.2 mmol), O-(4-(trifluoromethyl)benzoyl) oximes 2 (1.5 equiv., 0.3 mmol), and $K_2S_2O_5$ (2 equiv., 0.4 mmol), $FeCl_2$ (0.1 equiv., 0.02 mmol) and DCE (2 mL) was heated at 60 °C for 24 h in a sealed tube (10 mL). After the reaction finished, the reaction mixture was cooled to room temperature and diluted with dichloromethane (5 × 3 mL), and the organic layers were combined, washed with sat. aq. Na_2CO_3 (10 mL), dried (Na_2SO_4), and concentrated in vacuo. The residue was purified by column chromatography to afford the desired product.

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[20] CCDC-2073119 (**3 c**) contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data request/cif.

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